Characterizing Heterogeneous Property Graphs

A distributional and semantic measurement framework

Version 1.0 | 3 November 2026 | Measurement first; synthesis second

Executive overview

A property graph cannot be characterized adequately by node count, edge count, average degree and the number of labels. These statistics miss differences in connectivity, relationship roles, property schemas, value distributions and domain rules. This framework defines a *measurement profile* for an arbitrary existing directed property multigraph. A measured profile can subsequently become a target for synthetic data generation, but generation is not assumed or required.

Central distinction: *observed structure* (what the stored graph contains), *interpreted semantics* (what labels, properties and relationships mean), and *model-dependent inference* (latent roles, communities or inferred rules) must be reported separately. No graph statistic can infer domain meaning uniquely from topology or labels alone.

Layer	Core observations	What it cannot establish alone
Population	Node/edge counts; label-set frequencies; property-bearing entities	Domain meaning
Connectivity	In/out-degree distributions; components; neighborhoods; paths	Semantic roles
Typed relations	Frequencies of (source labels, edge type, target labels) and per-signature cardinalities	Validity of observed patterns
Properties	Presence-pattern frequencies, value distributions, list length, type variation	Meaning, units or correctness
Dependencies	Property association, topology–attribute coupling, relational dependencies	Causality
Declared semantics	Ontology, units, domains, allowed signatures, integrity rules	Complete world knowledge

Minimum characterization deliverable. Report a population summary; directed degree and typed-degree distributions; weak/strong components; sampled k-hop reachability; edge-signature frequencies; label-set frequencies; property-presence signatures; within-type property distributions; a small number of justified dependency tests; and explicit interpretation/coverage metadata. A single "complexity score" is not part of the core output.

Outputs. (i) A machine-readable *observation profile*, (ii) an optional semantic contract, (iii) an estimator/uncertainty manifest, and (iv) a human-readable analysis. For synthesis, append targets, priorities and tolerances; do not overwrite observed measurements with targets.

Two nonnegotiable safeguards. A metric is only comparable when its scope, graph projection, denominator and estimator agree. A metric is only semantically interpretable when the required domain assumptions have been supplied.

Contents

1 Object of study, scope and identifiability **3**
1.1 Directed property multigraph . . . 3
1.2 Measurement scope is part of the metric . . . 3
1.3 Levels of claim . . . 3

2 Structural characterization **3**
2.1 Scale, incidence, degree and density . . . 3
2.2 Connectivity, paths and local concentration . . . 4
2.3 Hierarchy is relation- and root-dependent . . . 4

3 Typed nodes, edges and relational roles **4**
3.1 Labels are observed classifications, not automatically types . . . 4
3.2 Complete edge signatures and mixing . . . 4
3.3 Cardinality as a conditional distribution . . . 5

4 Property schema and value heterogeneity **5**
4.1 Presence signatures . . . 5
4.2 Worked example: same marginals, different schemas . . . 5
4.3 Property values and representation . . . 5

5 Dependencies, semantic roles and constraints **5**
5.1 Attribute dependencies within a cohort . . . 5
5.2 Topology–attribute coupling . . . 6
5.3 Optional conceptual semantics contract . . . 6
5.4 Conceptual diversity is optional and domain-relative . . . 6

6 A characterization profile for any existing graph **6**
6.1 Metric contract . . . 6
6.2 Operational profile: compact JSON example . . . 7

7 Worked comparison: why distributions matter **7**
7.1 Three equally sized populations . . . 7
7.2 Same topology, different relationship roles . . . 8
7.3 Same semantics, different topology . . . 8

8 From measurement to synthetic generation **8**
8.1 Feasibility and trade-offs . . . 8
8.2 Practical pipeline . . . 8

9 Measurement and validation protocol **8**
9.1 Exact versus approximate . . . 8
9.2 Required tests before publication . . . 9
9.3 How to decide which metrics belong . . . 9

10 Implementation priorities and limitations **9**
10.1 Recommended measurement-first milestones . . . 9
10.2 Limits of generality . . . 9

1 Object of study, scope and identifiability

1.1 Directed property multigraph

Represent the stored graph as

$$G = (V, E, s, t, \tau, \ell, \pi_V, \pi_E), \tag{1}$$

where V and E are finite sets of node and relationship identities, $s, t : E \to V$ select source and target, $\tau : E \to \mathcal{R}$ is the single relationship type, $\ell : V \to 2^{\mathcal{L}}$ maps nodes to *sets* of labels, and π_V, π_E are partial key–value maps. Distinct relationships can share endpoints and types; self-loops are possible. This is a descriptive model of Neo4j's directed labeled property graph, not a prescriptive ontology.

A key may be *absent*; presence is different from a value used by an application to denote "unknown." Within Neo4j, a stored null property is not a third distinct state. The analysis must respect the platform's supported property types and list restrictions. In general, do not treat IDs, labels, values or relationship types as evidence of business meaning without a declared interpretation.

1.2 Measurement scope is part of the metric

Every result is indexed by a scope

$$\mathcal{S} = (V', E', d, m, w, r, \theta), \tag{2}$$

where $V' \subseteq V$ and $E' \subseteq E$ identify a selected subgraph; d chooses directed or undirected traversal; m chooses edge multiplicity handling; w specifies weights (if any); r is a root/sample policy; and θ records filters, time windows and preprocessing. Example: undirected clustering after collapsing parallel edges is a *different quantity* from motif counts in the original directed multigraph.

1.3 Levels of claim

1. **Observed:** counts, label sets, value presence and exact finite-population frequencies.
2. **Derived under a projection:** shortest paths, clustering, centrality and modularity (requires a projection, algorithm or partition).
3. **Model-derived:** latent subtypes, inferred ontology, estimated semantic distances and missing rules. Report model and uncertainty.

The first level can be measured without ontology. The third cannot be established uniquely from raw graph data. The objective is *characterization*, not pretending those levels are interchangeable.

2 Structural characterization

2.1 Scale, incidence, degree and density

Let $n = |V'|$, $m = |E'|$. Define

$$k_{\text{out}}(v) = |\{e \in E' : s(e) = v\}|, \qquad k_{\text{in}}(v) = |\{e \in E' : t(e) = v\}|. \tag{3}$$

Loops contribute one to each degree. Consequently $\sum_v k_{\text{out}} = \sum_v k_{\text{in}} = m$ and average total incidence is $2m/n$. Record the full joint empirical distribution

$$\widehat{p}_{\mathcal{S}}(a, b) = \frac{1}{n} \sum_{v \in V'} \mathbf{1}[k_{\text{in}}(v) = a, k_{\text{out}}(v) = b], \tag{4}$$

plus percentiles, isolate fraction, maximum, zero inflation and the dependence of in/out degree. A power-law exponent is *not* a default metric: fit it only with a documented tail range and goodness-of-fit diagnostic.

For a loopless directed *simple* graph, density is $m/[n(n-1)]$. For a multigraph use

$$\rho_{\text{pair}} = \frac{|\{(u, v) : u \neq v, \exists e,\ s(e) = u, t(e) = v\}|}{n(n-1)}, \quad \mu = \frac{m_{\text{nonloop}}}{\#\text{distinct ordered nonloop pairs}}. \tag{5}$$

The pair occupancy $\rho_{\text{pair}} \in [0, 1]$ and the multiplicity $\mu \geq 1$ (when defined) separate the presence of a connection from parallel relationships. Report loop count separately. An "edge-count density" can exceed one on a multigraph and should not be called a probability.

2.2 Connectivity, paths and local concentration

Count weakly connected components (ignore direction) and strongly connected components (respect direction); report size distributions and largest-component share. For directed paths, specify reachability and exclude unreachable node pairs from finite distances, while reporting their fraction. Define

$$B_d(v) = |\{u : \mathrm{dist}(v,u) = d\}|, \qquad R_k(v) = \sum_{d=0}^{k} B_d(v). \tag{6}$$

The distribution of $R_k(v)$ over sampled roots is more stable and actionable for query characterization than one maximum "depth." An expansion ratio B_{d+1}/B_d is defined only when $B_d > 0$. Differentiate distinct reachable *nodes* from number of matching *paths*; the latter may grow much faster.

Clustering requires a declared triangle convention: ordinary undirected clustering is computed on a simple undirected projection, while directed triad or motif counts retain orientation. Effective diameter should specify a quantile (e.g. 90th) and the component/pair sampling policy. Community modularity is a *partition-dependent* result, not an invariant of the graph alone. Record the partition algorithm, random seed and resolution.

2.3 Hierarchy is relation- and root-dependent

Arbitrary graphs have neither a unique root nor a unique level decomposition. For a chosen relation subset E_h, one may analyze DAG depth when the subset is acyclic; otherwise use BFS layers or a condensed strongly-connected-component DAG. A node's "depth" means nothing without specifying root selection, direction and relation filter. This is essential for mixed domains where one relation forms a hierarchy while others form cycles.

3 Typed nodes, edges and relational roles

3.1 Labels are observed classifications, not automatically types

Record (i) each label's prevalence, (ii) the full label-set distribution $p(L)$ with $L = \ell(v)$, (iii) cardinality $|L|$, and (iv) co-occurrence and implication rates. For labels A, B, report

$$\mathbb{P}(B \mid A) = \frac{|\{v : A, B \in \ell(v)\}|}{|\{v : A \in \ell(v)\}|}. \tag{7}$$

A high implication rate is an observed regularity, not proof that A is formally a subtype of B. A canonical type $T(v)$ is *optional* and must be supplied by a mapping, selected explicitly by an analyst, or inferred with a stated model. Never silently replace overlapping label sets with one arbitrary label.

For a categorical variable Z with empirical proportions p_z, define Shannon diversity and effective diversity

$$\mathrm{H}(Z) = -\sum_{z} p_z \log_2 p_z, \qquad D_{\mathrm{eff}}(Z) = 2^{\mathrm{H}(Z)}. \tag{8}$$

These measure balance, not conceptual distance. Two equally distributed taxonomies can have identical entropy despite radically different meanings.

3.2 Complete edge signatures and mixing

Without a canonical type, the directly observable signature is

$$\sigma(e) = (\ell(s(e)), \tau(e), \ell(t(e))). \tag{9}$$

A declared type map permits the simpler (T_s, R, T_t). Measure signature frequencies $p(\sigma)$ and conditional mixing $p(L_t \mid L_s, R)$. Also report type-specific relationship counts and degree distributions

$$K^{\mathrm{out}}_{r,L}(v) = \sum_{e : s(e) = v} \mathbf{1}[\tau(e) = r, \ell(t(e)) = L], \tag{10}$$

with the analogous incoming quantity. The vector $\mathbf{K}(v)$ across $(r, L, \text{direction})$ describes a node's *relational role*, even among nodes with identical labels and properties. Measure its distribution conditional on label set or declared type.

Careful denominators. "Signature sparsity" is undefined until the candidate universe is fixed. The share of *observed* signatures among all combinatorially possible triples differs from the share among *declared allowed* triples. Report both only when their respective universes are defined. A relationship's existence cannot by itself demonstrate that it is semantically valid.

3.3 Cardinality as a conditional distribution

A mean of two outgoing `DEPARTS_FROM` edges per flight could conceal one half with zero and the other half with four. Therefore report

$$\mathbb{P}\big(K_{r,L}^{\text{out}} = k \mid T = t\big) \tag{11}$$

including zero counts, not just nonzero participants. Cardinality constraints (exactly one, at most one, many) belong to an optional semantic contract; empirical cardinality violations need that contract.

4 Property schema and value heterogeneity

4.1 Presence signatures

For entity type or cohort t define its observed candidate-key universe $\mathcal{K}_t$ and $X_{vp} = \mathbf{1}[p \in \text{dom}(\pi_V(v))]$. The presence signature $\mathbf{X}_v \in \{0,1\}^{|\mathcal{K}_t|}$ gives a complete observed property schema. Measure (a) each $q_p = \mathbb{P}(X_p = 1 \mid t)$, (b) property count $Q = \sum_p X_p$, (c) frequency of complete presence signatures, and (d) entropy $\mathrm{H}(\mathbf{X} \mid t)$. Calculate the same for edge properties within relationship-signature cohorts.

The identity

$$\mathrm{H}(\mathbf{X} \mid t) \le \sum_{p \in \mathcal{K}_t} \mathrm{H}(X_p \mid t) \tag{12}$$

exposes dependence: individual property-presence entropy cannot substitute for the joint distribution. The gap (total correlation) is

$$\mathrm{TC}(\mathbf{X} \mid t) = \sum_p \mathrm{H}(X_p \mid t) - \mathrm{H}(\mathbf{X} \mid t) \ge 0. \tag{13}$$

Interpretation: a large gap indicates dependence among presence indicators, not necessarily a complex ontology. Adding keys absent from every entity changes neither measure; keys and cohort policy must nevertheless be recorded for reproducibility.

4.2 Worked example: same marginals, different schemas

Consider four nodes with properties A and B. In population I, the signatures are $(1,0),(1,0),(0,1),(0,1)$. In population II, they are $(0,0),(0,0),(1,1),(1,1)$. Both have $\mathbb{P}(A=1) = \mathbb{P}(B=1) = 0.5$, $\mathbb{E}Q = 1$, and joint entropy $\mathrm{H}(A,B) = 1$ bit. Yet the relationship between keys is opposite: presence is mutually exclusive in I and perfectly co-occurring in II. *Entropy alone does not identify the dependency.* Report the complete signature distribution or an association sign together with total correlation.

A third population with each of $(0,0),(0,1),(1,0),(1,1)$ once has identical marginals but $\mathrm{H}(A,B) = 2$ bits and $\mathrm{TC} = 0$. The difference is experimentally verifiable and shows why a scalar "schema complexity" is inadequate.

4.3 Property values and representation

Only compare values conditional on *presence*, a declared interpretation and a compatible data type. For categorical values, report frequencies, distinct count and entropy; for numeric values, quantiles, robust dispersion, heavy tails and multimodality; for strings, length/format distributions; for lists, length and element-type distributions. Mixed-type keys require a type-mixture profile. Distinguish domain sentinels (such as `UNKNOWN`) from absent properties.

Avoid comparing raw differential entropies across physical units; they change with unit scaling. For numeric differences across cohorts, use quantile distances after justified unit conversion or Wasserstein distances in the same units. High cardinality identifiers should be flagged: their entropy usually measures identity, not domain complexity. Record whether a property is excluded as an ID, timestamp, free-text field or technical metadata.

5 Dependencies, semantic roles and constraints

5.1 Attribute dependencies within a cohort

For jointly observed A, B, mutual information is

$$\mathrm{I}(A; B \mid T = t) = \sum_{a,b} p(a,b \mid t) \log_2 \frac{p(a,b \mid t)}{p(a \mid t)p(b \mid t)}. \tag{14}$$

This describes statistical dependence, not causality. Its estimation requires discrete values or explicitly chosen estimators; naive discretization, sparse categories and unique IDs can bias results. Report sample size, estimator,

missingness strategy and (preferably) a permutation or bootstrap reference. A global association may vanish or reverse within subtypes, so analyze *conditional* distributions where justified.

5.2 Topology–attribute coupling

A node's attribute profile and role vector can be coupled even within one declared type:

$$p(\mathbf{X}, \mathbf{Y}, \mathbf{K} \mid T = t). \tag{15}$$

Examples: job title correlates with reporting degree; aircraft category correlates with operational relationships. Compare degree/cardinality distributions between property-defined cohorts, or estimate $\mathrm{I}(Y; K \mid T)$ with explicit discretization and controls. For relational properties, define a sampled edge-level table $((Y_s, Y_t), R)$ and assess conditional dependence; avoid treating edges sharing endpoints as independent observations in uncertainty calculations.

5.3 Optional conceptual semantics contract

A database is not a self-describing ontology. To evaluate semantics rather than only observed regularities, define a contract $\mathcal{O}$ containing:

- Named concepts and optional subsumption/role relationships; explicit mapping from label sets to concepts.
- Property meanings, units, permissible domains, value normalization and identity conventions.
- Allowed (source concept, relation, target concept) signatures, direction and multiplicity.
- Local and relational rules: exclusivity, implications, functional dependencies, temporal conditions and graph-wide invariants.

Given rules C_j, report violation count and rate with a documented denominator N_j of *eligible tests*:

$$v_j = \frac{\#\{\text{eligible tests failing } C_j\}}{N_j}, \qquad N_j > 0. \tag{16}$$

Rule-violation frequency measures conformity, *not* complexity of the rule system. A rule-count score cannot measure semantic difficulty reliably because logically equivalent constraints can be written in different forms. Report rule language, logical coverage and dependencies instead of an ungrounded scalar.

5.4 Conceptual diversity is optional and domain-relative

If an ontology or a justified concept embedding is supplied, report semantic distances $d_{\mathrm{sem}}(t_i, t_j)$ and type-weighted dispersion, e.g.

$$D_{\mathrm{concept}} = \sum_{i<j} 2 p_i p_j d_{\mathrm{sem}}(t_i, t_j). \tag{17}$$

This quantity is *not canonical*: changing the ontology, distance, granularity or type mapping changes its value. Do not compare it across domains without aligned semantics. In the ontology-free case, report it as *not available*, not zero.

6 A characterization profile for any existing graph

6.1 Metric contract

A metric record should contain more than a number:

$$M = (\text{name}, \text{definition}, \text{scope}, \text{value}, \text{estimator}, \text{uncertainty}, \text{status}). \tag{18}$$

The status distinguishes exact, estimated, not-applicable, not-measured, and ontology-required. Record source snapshot/version and all projections. Approximate metrics should include sampling strategy, sample size, seed and confidence intervals where meaningful; uncertainty is about estimation, not about the intrinsic finite graph.

Metric family	Quantity and required scope	Computation	Interpretation / caveat
Population	n, m; label-set distribution; edge types	Exact scan or query	Straight counts; labels need not be semantic types.

Metric family	Quantity and required scope	Computation	Interpretation / caveat
Degrees	$p(k_{in}, k_{out})$; typed role vector $p(\mathbf{K} \mid L)$	Exact grouping; tails can be sampled	Multiplicity, self-loops and cohort definition matter.
Pair connectivity	Pair occupancy, multiplicity	Distinct endpoint pairs	Separate pair density from multiedge intensity.
Components	WCC/SCC size distributions	Graph algorithm	Report directedness and isolates.
Paths	$p(R_k)$, distance quantiles, unreachable share	BFS over root samples	Root policy and path semantics essential.
Local topology	Triangle/motif and clustering distributions	Defined projection + algorithm	No unique directed multigraph clustering coefficient.
Communities	Membership sizes, modularity	Selected algorithm/partition	Algorithm-, resolution-, seed-dependent.
Relation signatures	$p(L_s, R, L_t)$	Exact edge scan	Label-set combinations can be large.
Schema signatures	$p(\mathbf{X} \mid L)$; $p(Q \mid L)$	Exact key-mask grouping	May require hashing/sparse representation.
Values	Categorical counts; numeric quantiles, type mix	Type-specific estimators	IDs and units must be handled explicitly.
Dependencies	Presence co-occurrence, conditional MI, role–attribute association	Targeted estimators	Not causal; multiple-testing and bias concerns.
Semantics contract	Rules, constraints, violation rates	User-supplied contract + validation	Cannot be inferred unambiguously from raw data.

6.2 Operational profile: compact JSON example

```
{
  "source": {"system":"neo4j", "snapshot":"2026-11-03", "scope":"all"},
  "definitions": {"direction":"directed", "multiplicity":"retain",
                  "cohorts":"exact_label_set"},
  "observed": {"nodes":100000, "relationships":600000,
               "label_sets": {"Flight":45000, "Airport":8000}},
  "metrics": [{"id":"reachability.k3", "scope":"all",
               "estimator":"BFS uniform-root sample", "sample_size":2000,
               "status":"estimated", "value":{"p50":18,"p90":140}}],
  "semantics": {"contract":"none", "conceptual_diversity":"unavailable"}
}
```

This is an *illustrative excerpt*, not a validated schema, and the example quantities do not imply that a real dataset was analyzed.

7 Worked comparison: why distributions matter

7.1 Three equally sized populations

Construct three synthetic populations of eight nodes, each with a single label `Device`, two candidate properties A, B, and identical property marginals $p(A = 1) = p(B = 1) = 1/2$. Repeat each four-record pattern twice:

Population	Four patterns	$H(A, B)$	$\mathrm{TC}(A, B)$	Inference
Exclusive	$10, 10, 01, 01$	1 bit	1 bit	Never co-present
Coupled	$00, 00, 11, 11$	1 bit	1 bit	Always equal
Independent	$00, 01, 10, 11$	2 bits	0 bits	Empirically independent

All three have two keys, the same marginal presence, the same per-key entropy and the same label count. The first two also have the same joint entropy and total correlation, but opposite co-occurrence structure. Thus *no small scalar metric inventory can fully replace the underlying distributions.* Preserve signatures and targeted contingency tables for interpretability.

7.2 Same topology, different relationship roles

If a graph's untyped edges form the same directed topology in two databases, but one assigns outgoing `MANAGES` edges predominantly to `Manager` nodes while the other randomizes these edge types, structural metrics are unchanged. Yet $p(K^{out}_{\texttt{MANAGES},\texttt{Employee}} \mid \ell(v))$ changes. A typed role profile detects the difference without an ontology; explaining which assignment is *correct* still requires an external semantic contract.

7.3 Same semantics, different topology

Conversely, keep all labels, relationship signatures and property tables constant while rewiring edges within allowed signatures. Component count, $p(R_k)$, community separation and clustering may change. This motivates reporting structural and semantic layers together instead of collapsing them into "heterogeneity."

8 From measurement to synthetic generation

Characterization is the primary operation: $G \mapsto \widehat{\mathcal{P}}(G)$. Synthesis is optional: $(\mathcal{P}^*, \mathcal{O}, \eta) \mapsto G^*$, where $\mathcal{P}^*$ is a target profile, $\mathcal{O}$ contains hard semantic rules, and η controls the random seed and generation algorithm. The measured profile of G^* must be compared with $\mathcal{P}^*$; generating data does not guarantee matching it.

8.1 Feasibility and trade-offs

Targets are coupled: $\mathbb{E}k_{out} = m/n$; component structure bounds reachability; typed cardinalities restrict degree sequences; a tree relation has no cycles; all constraints restrict possible joint property signatures. Some targets are exact, others approximate, and some may be mutually infeasible. Record hard constraints, soft loss terms, weights and tolerances. One useful problem formulation is

$$G^* \in \arg\min_{G \in \mathcal{G}(\mathcal{O})} \sum_i w_i D_i\big(\widehat{\mathcal{P}}_i(G), \mathcal{P}^*_i\big), \tag{19}$$

where $\mathcal{G}(\mathcal{O})$ is the set of graphs satisfying mandatory semantic constraints. The distances D_i are chosen per metric: exact count errors, total variation for discrete frequencies, Wasserstein for compatible numeric distributions, or direct validation for hard rules. This is an engineering optimization objective, not a guarantee that every target is achievable.

8.2 Practical pipeline

1. **Measure:** extract finite-population counts and distribution estimates from an existing graph, including scope/uncertainty.
2. **Interpret:** attach an optional semantic mapping and explicitly authored constraints; keep inferred assumptions marked.
3. **Select targets:** choose which measured quantities are to be preserved and at what tolerance.
4. **Synthesize:** generate entities, property-presence signatures and conditional values; then generate relationships respecting typed cardinalities; optionally refine topology.
5. **Validate:** remeasure the generated graph using the *same* metric definitions and sampling policy; compare distributions and rule violations.

A heterogeneous degree-corrected block model can provide a starting connectivity model; it does not by itself reproduce triangles, arbitrary functional dependencies or temporal consistency. These require separate mechanisms or constrained refinement.

9 Measurement and validation protocol

9.1 Exact versus approximate

Exact counts and grouped frequencies can be computed through scans; full shortest-path distributions, complex motifs and joint high-dimensional property distributions may be prohibitively expensive. Use stratified cohorts or sampled roots and report sampling weights. When a sample misses rare high-degree hubs or rare labels, broad confidence intervals do not necessarily correct sampling *bias*. State exact computational scope, projection and exclusions.

9.2 Required tests before publication

1. **Algebraic identities:** verify $\sum k_{in} = \sum k_{out} = m$; $0 \leq \rho_{pair} \leq 1$; normalized frequency vectors sum to one.
2. **Counterexamples:** confirm the exclusive, coupled and independent property patterns give the values in Section 8; verify label-count and degree controls cannot distinguish them.
3. **Representation invariance:** renaming node IDs must not change structural metrics; relabeling classes consistently must not change diversity values; changing units must not be interpreted as changed physical dispersion.
4. **Projection sensitivity:** test directed vs undirected and multigraph vs simple projections; require results to carry scope changes.
5. **Estimator diagnostics:** compare sampled BFS reachability to exact measurements on small graphs; test sample-size convergence and random-seed stability.
6. **Semantics boundary:** absent ontology must return *unavailable* for conceptual distance and rule conformity, not fabricated numbers.
7. **Generation fidelity:** compare per-cohort distributions, rare-signature coverage, dependencies and constraint violations; provide unachieved targets explicitly.

9.3 How to decide which metrics belong

A candidate metric should be kept only if (1) its mathematical object is defined; (2) its scope/denominator is unambiguous; (3) it distinguishes relevant graph families not already captured by simpler outputs; (4) its estimator and cost are known; and (5) its interpretation does not overclaim semantic meaning. This framework intentionally retains *distributions and contracts* instead of promoting dozens of weakly justified one-number scores.

10 Implementation priorities and limitations

10.1 Recommended measurement-first milestones

1. Scan/stream Neo4j nodes and edges into exact label-set, relationship-type, property-key and incidence summaries.
2. Compute typed edge-signature and per-cohort cardinality distributions; audit self-loops and parallel edges.
3. Compute component-size distributions and sampled neighborhood expansion with explicit graph projection settings.
4. Characterize property-presence signatures and per-key value distributions for node *and edge* cohorts.
5. Add targeted association measures, then optional semantic contracts and rule validators.
6. Implement comparison between two observed graphs; only then use the same profile as a target for optional synthesis.

10.2 Limits of generality

This method applies to arbitrary finite property graphs at the *observational* level. It cannot guarantee complete meaning, causal structure, valid ontology, intended temporal semantics, or statistical identifiability of latent subclasses without additional domain assumptions. Huge label combinations or property-signature spaces require sparse storage, rare-category grouping (with provenance), or approximation. Cross-graph comparisons require aligned scope and often aligned ontologies. Query performance additionally depends on indexes, execution plans, hardware and workload; it should be benchmarked as a separate operational dimension, not treated as an intrinsic graph property.

References and precise provenance

1. Neo4j. *Graph database concepts*. Property-graph semantics and multi-label nodes. https://neo4j.com/docs/getting-started/appendix/graphdb-concepts/

2. Neo4j. *Graph Data Science: Weakly Connected Components*. Direction conventions. https://neo4j.com/docs/graph-data-science/current/algorithms/wcc/

3. Neo4j. *Graph Data Science: Triangle Count*. Undirected triangle-count convention in GDS. https://neo4j.com/docs/graph-data-science/current/algorithms/triangle-count/

4. Shannon, C. E. (1948). *A Mathematical Theory of Communication*. Bell System Technical Journal, 27, 379–423, 623–656.

5. Newman, M. E. J. (2003). *The Structure and Function of Complex Networks*. SIAM Review 45(2), 167–256.

6. Holland, P. W., Laskey, K. B., Leinhardt, S. (1983). *Stochastic blockmodels: First steps*. Social Networks 5(2), 109–137.

7. Karrer, B., Newman, M. E. J. (2011). *Stochastic blockmodels and community structure in networks*. Physical Review E 83, 016107.